

IBRAHIM ROGERS

MECHANICAL ENGINEERING | TEST SYSTEMS & METROLOGY | HARDWARE DEVELOPMENT

B.S. Mechanical Engineering, Northeastern University · Expected May 2028
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SELECTED WORK

01	Liquid Valve Endurance Tester & Leak-Test Fixture	Parker Hannifin · 2026
02	Refreshable Braille Tactile Display	Forge · 2026
03	Vision-Guided Pan/Tilt Tracking System	Independent · 2025-26
04	Gamma Shielding Test System	Melagen Labs · 2024
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Liquid Valve Endurance Tester & Leak-Test Fixture

Quality & Sustaining Engineering Co-op — Parker Hannifin, Precision Fluidics · Hollis, NH · May 2026 – Present

OVERVIEW

Context: Precision fluid-control valves for medical and life-science systems. My work spans test-system design, fixture design, and root-cause investigation on returned product.

My Role: Requirements definition and hydraulic architecture for a closed-loop endurance tester; mechanical design of a supplier-side leak-test fixture; lead on a corrosion root-cause investigation.

PROJECT A — CLOSED-LOOP DI-WATER VALVE ENDURANCE TESTER

Converted conflicting stakeholder inputs from quality, manufacturing, and engineering into a single defined test-system scope, then developed the architecture and proof-of-concept P&ID to match.

Test capacity	30–60 valves cycled simultaneously
Flow per valve / aggregate	0.5 L/min / 15–30 L/min
Operating pressure	120 psig
Fluid temperature control	±5°F of ambient
Working fluid	DI water (continuous recirculation)

Sized pump, chiller, reservoir, manifold, and instrumentation, and integrated pressure control, relief protection, filtration, and a chiller bypass loop for thermal management under continuous recirculation. Architecture reserves interfaces for future multi-fluid capability without redesigning the downstream test rack.

PROJECT B — PRESSURE-DECAY LEAK-TEST FIXTURE

Designed a fixture in Autodesk Inventor for supplier-side incoming inspection of a molded fluid-control component.

Leak sensitivity requirement	< 0.1 sccm at the orifice
Trapped volume	< 0.5 cc
Method selected	Pressure decay (over make-up flow)
Seal architecture	Guarded three-port
Probe sizing basis	Worst-case tolerance (±0.001 in) + MMC/LMC

DESIGN REASONING

Pressure decay was selected over steady-state make-up flow because commercially available flow instrumentation could not resolve the sub-0.1 sccm requirement at acceptable cost. Locating and sealing functions were separated so the probe seals against the orifice without over-constraining a molded part with real dimensional variation.

PROJECT C — ORIFICE CORROSION ROOT-CAUSE INVESTIGATION

Lead investigator on orifice corrosion affecting 70%+ of returned valve bodies. Designed isolation experiments across supplier, stock, cleaning, and floor-process variables to identify the contamination source and validate mitigation. Separately, investigated 500+ contamination and returned-product failures through teardown, comparative inspection against conforming units, and manufacturing-process review, authoring customer-facing failure-analysis reports.

Refreshable Braille Tactile Display

Hardware Product Lab — Forge, Northeastern University · Jan 2026 – Jun 2026

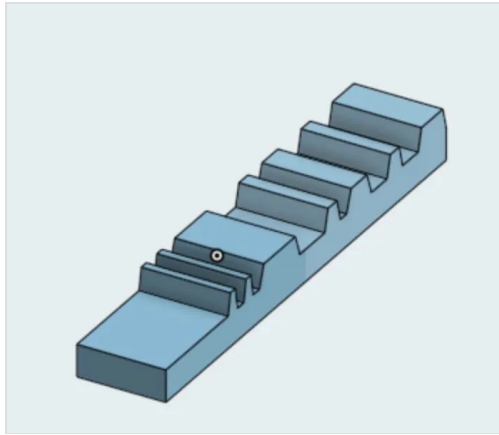
OVERVIEW

Challenge: Design a low-cost, mechanically actuated Braille cell that survives thousands of actuation cycles while holding pin position accurately enough to stay tactilely readable.

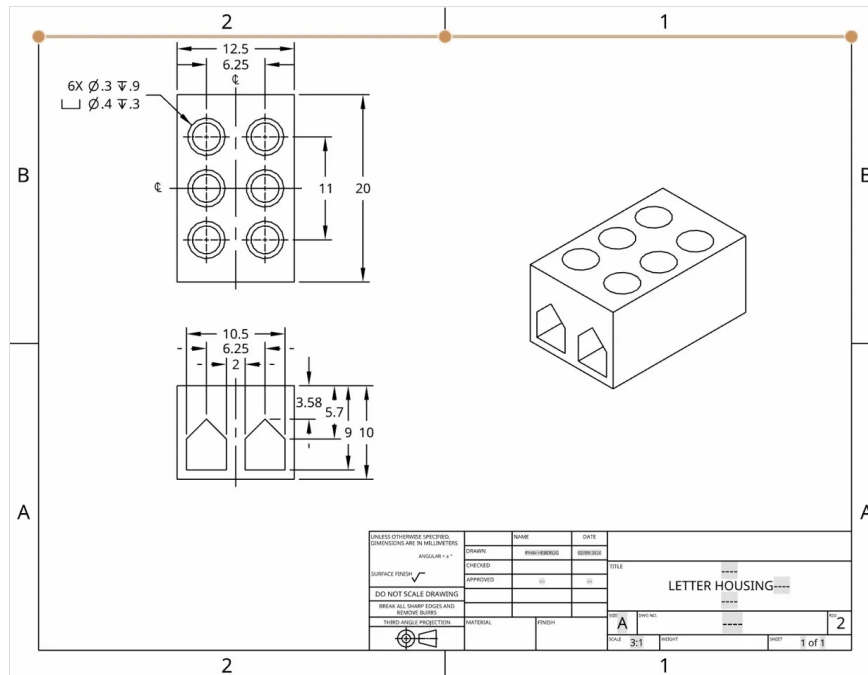
My Role: Mechanical design of the actuation mechanism, clearance and tolerance-stack iteration, prototype fabrication, and cycle testing.

Approach: Bevel/spur gearing with rack actuation converts motor rotation into linear key motion, packaged within a $31 \times 49 \times 19$ mm envelope. Modeled in Onshape; clearances refined iteratively against physical failure modes rather than analysis alone.

MECHANISM DESIGN

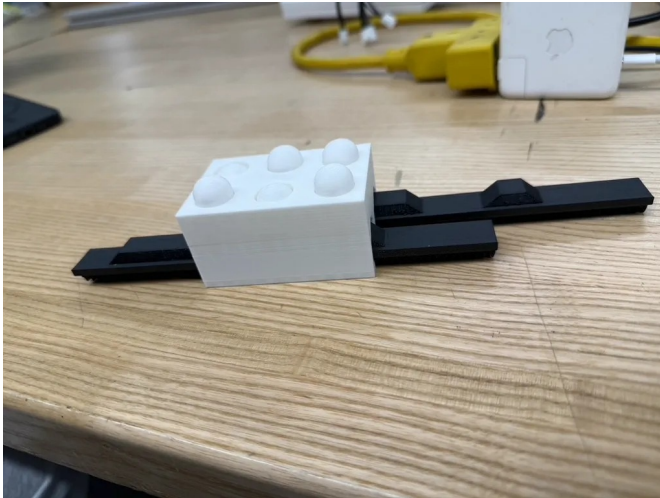


Onshape CAD — stepped rack. Each step drives an individual pin through vertical translation as the rack advances. Geometry was tuned for low friction while keeping pin positional deviation within the readable tactile envelope.

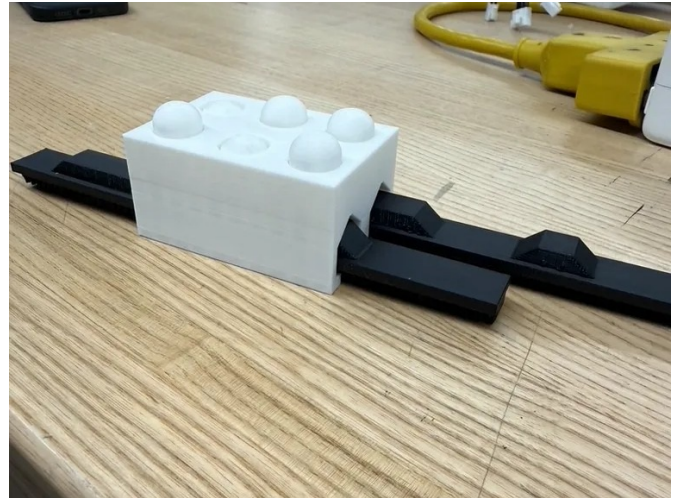


Letter housing detail drawing. 2×3 dot pattern on a 6.25 mm pitch in a $20 \times 12.5 \times 10$ mm housing. Six $\varnothing 0.3$ mm pin bores, 0.9 mm deep, each with a $\varnothing 0.4$ mm $\times$ 0.3 mm counterbore as a lead-in for pin entry. Drawing prepared by teammate R. Heberlig; I owned the actuation mechanism, clearance scheme, and test iteration described on this page.

PROTOTYPING & ITERATION



Assembled prototype — side view. White housing carries the pin actuation mechanism and rides on the black guide rails, which provide lateral constraint. This view shows rail engagement and the travel available to the rack.



Same prototype — front three-quarter view. Shows the six-dot face and the housing-to-rail interface where side-load binding originated. Lead-in chamfers on the rails were added at this interface to ease pin entry.

KEY PROBLEM — TOLERANCE STACK-UP UNDER CYCLIC LOADING

Early prototypes jammed when actuated under side load. Failure tracking isolated two compounding causes:

- Guide-channel clearance too tight at 0.1 mm nominal — insufficient margin once print tolerance and side deflection stacked.
- Dimensional creep in PLA after sustained cycling, drifting clearances further out of spec.

Fix: Opened guide clearance to 0.15 mm, added lead-in chamfers, and moved to PETG. **Result:** jam rate fell from roughly 15% to under 2% across 2,000+ test cycles — close to an order-of-magnitude reduction in failure rate.

WHAT I TOOK FROM IT

- **Clearance is not just a manufacturing number.** It has to absorb print tolerance *and* accumulated wear, not just nominal fit.
- **Material choice is a lifecycle decision.** PLA was fine for form checks and wrong for cyclic loading; that only showed up in testing.
- **Track failures, don't guess.** Logging jam events by condition pointed at side-load binding specifically, which is what made the fix targeted instead of speculative.
- **Design for assembly.** Consolidating cam and guide features cut part count from 12 to 8 and simplified the assembly sequence.
- **Cross-domain constraint.** Worked with the electrical side to keep actuation force inside servo torque limits while still meeting tactile force requirements.

Vision-Guided Pan/Tilt Tracking System

Independent Project · Dec 2025 – Present

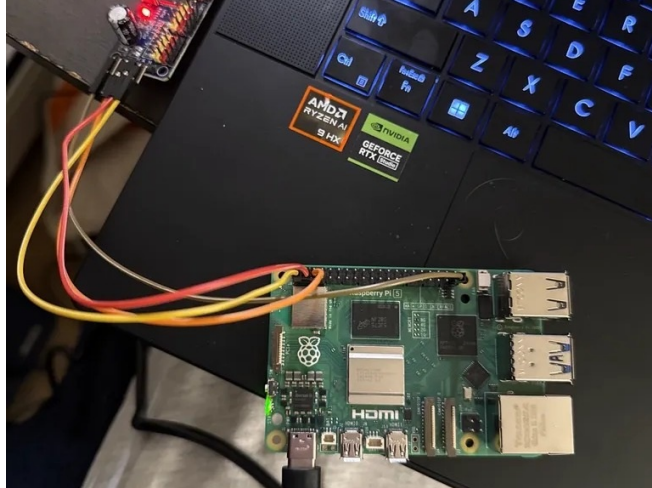
OVERVIEW

Goal: Build a closed-loop target tracking platform — mechanical gimbal, electronics, and control software — that holds a marker centered in frame and recovers on its own when the target is lost.

System: Raspberry Pi 5, camera module, PCA9685 servo driver, dual-servo pan/tilt gimbal, OpenCV ArUco detection, Python control logic, Flask live video stream.



Pan/tilt gimbal. Custom two-axis 3D-printed housing with direct-drive servo coupling — no gear train, which removes backlash from the positioning path.



Electronics integration. Raspberry Pi 5 driving the PCA9685 servo controller over I²C, with servo power kept on a separate rail from compute power to keep actuation transients off the Pi.

MECHANICAL DESIGN DECISIONS

- **Backlash:** direct-drive servo coupling with a press-fit interface, eliminating gear train slop from the control loop.
- **Center of mass:** camera placement chosen to reduce servo torque load and improve response during fast target movement.
- **Rigidity:** gimbal frame reinforced to limit flex under rapid acceleration so the camera stays aligned during dynamic tracking.
- **Cable routing:** integrated to avoid wire interference across the full pan/tilt range.

INTEGRATION DEBUG — CAMERA NOT ENUMERATING

PROBLEM

No frame acquisition and no video feed. From the outside the whole system looked dead, with no obvious indication whether the fault was mechanical, electrical, or software.

PROCESS

- Used `raspistill` to separate camera hardware faults from software faults.
- Verified I²C device enumeration to confirm the Pi was seeing devices on the bus.
- Found improper CSI ribbon seating compounded by outdated camera firmware — two independent faults presenting as one symptom.
- Updated firmware and added error handling in Python for graceful frame-drop recovery.

LESSON

Integration bugs concentrate at interface boundaries. Isolating with diagnostic tools beat guessing at the failure location, especially with two faults stacked.

CONTROL IMPLEMENTATION

- PID loop on pan and tilt axes to smooth servo motion and suppress oscillation around the target; gains tuned empirically on hardware.
- Automated re-acquisition via spiral search when the marker leaves frame or is occluded.
- Error handling for dropped frames, I²C communication failures, and servo limit violations.

WHAT I TOOK FROM IT

- **Stage the integration.** Camera alone, then servos alone, then detection, then full loop. Bringing it up in stages saved hours over debugging everything at once.
- **Textbook PID isn't enough.** Real servo dynamics, friction, and loop latency meant the gains had to come from testing, not from the equations.
- **Cross-domain debugging is the actual skill.** Mechanical slop, power distribution, and software timing all produced similar-looking tracking errors.

Gamma Shielding Test System

Co-Founder & COO — Melagen Labs, VC-backed aerospace materials startup · Jan 2024 – Sep 2024

OVERVIEW

Challenge: Validate the shielding effectiveness of a melanin-HDPE composite against an aluminum reference using gamma radiation, with limited lab equipment and a startup budget.

My Role: Co-designed the test fixture, developed the experimental protocol for repeatable measurement, ran the testing, and performed the statistical analysis.

Key Result: Attenuation comparable to aluminum at roughly one-third the density, supported by ANOVA.

TEST SYSTEM & SETUP



Complete test system. Fixed-geometry fixture holds a repeatable Cs-137 source-to-detector distance. A modular sample holder allows quick swaps between composite thicknesses without disturbing the measurement geometry.

DESIGN REQUIREMENTS

- **Repeatability:** fixed source-to-detector positioning to minimize geometric variation between runs — the dominant error source in this measurement.
- **Modularity:** quick-change sample mounting to test multiple thicknesses and compositions efficiently.
- **Safety:** Cs-137 source containment and shielding per lab safety protocol.
- **Low-cost fabrication:** built from lab-available materials within startup budget.

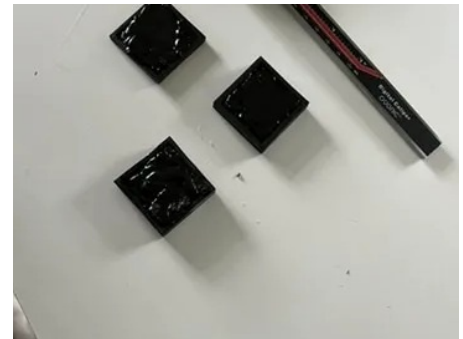
SAMPLE PREPARATION & MEASUREMENT



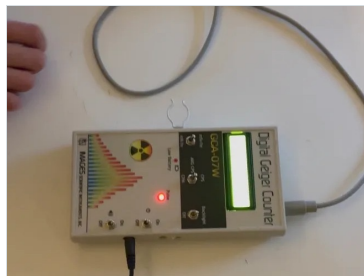
Material preparation. Melanin powder measured to a controlled mass so concentration stayed consistent batch to batch.



Sample fabrication. Composite cast in aluminum pans for consistent thickness, under controlled material-handling protocol.



Test specimens. Composite plates at varying thickness, measured and tested against aluminum references.



Radiation measurement. Calibrated Geiger counter at fixed distance from the shielded source, logging count rate for each shielding condition against an aluminum baseline.

ENGINEERING PROCESS

- **Fixture design for repeatability:** hand-positioning produced unacceptable run-to-run measurement scatter. Adding fixed mechanical alignment features cut that variation substantially and was the single biggest improvement to data quality.
- **Rapid iteration under constraints:** when the first fixture drifted after repeated sample changes, redesigned the mounting within days using available shop tools rather than waiting on ideal equipment.
- **Statistical validation:** ANOVA across material types and thicknesses, with regression modeling of shielding effectiveness against melanin concentration.
- **Uncertainty analysis:** documented contributions from source activity calibration, detector efficiency, and geometric positioning so the results were defensible to outside readers.
- **Technical communication:** translated the data into engineering presentations used in fundraising.

WHAT I TOOK FROM IT

- **Fixture design drives data quality.** In experimental work, repeatability is the whole game — small mechanical improvements to alignment had far more impact on confidence in the result than anything downstream in the analysis.
- **Constraints force decisions.** Limited budget and equipment meant choosing what actually mattered to measure well, rather than trying to control everything.

Technical Skills

Tools and methods I can speak to in detail and have applied on the work in this portfolio

CAD & Design

Autodesk Inventor · SolidWorks · Onshape · AutoCAD
GD&T (MMC/LMC) · worst-case tolerance stack-up · interference analysis · fixture design · engineering drawings · P&ID development

Programming & Analysis

Python (OpenCV, pandas) · C++ · MATLAB · Excel
ANOVA / regression · I²C and PWM interfaces · Raspberry Pi · Arduino · hardware-software integration

Quality & Validation

Failure analysis · root-cause investigation · FMEA · dimensional inspection & metrology · experimental test design · nonconformance investigation · validation and test documentation

Fluid & Test Systems

Hydraulic/pneumatic architecture · pressure regulation & relief · closed-loop recirculation · pump and chiller sizing · manifold design · pressure-decay and flow-based leak test methods · test-stand architecture

Manufacturing & Prototyping

FDM 3D printing (PLA, PETG) · SLA/resin printing · rapid prototyping · mechanical assembly · fixture fabrication · design for manufacturing · design for assembly

Engineering Communication

Customer-facing failure-analysis reporting · design reviews · cross-functional requirements definition · vendor and supplier coordination · technical presentations to non-technical audiences

ADDITIONAL PROJECT EXPERIENCE

Assistive Sweeper Robot

Cornerstone II, London · Jan-Apr 2025

Co-designed an RC sweeper in SolidWorks, working drivetrain geometry against motor torque and ground clearance constraints. Iterated through a rapid prototyping sprint for operator usability and assembly.

Autonomous Maze Robot

Cornerstone I, Oakland · Sep-Dec 2024

Programmed sensor-based autonomous navigation logic. Diagnosed drivetrain binding and electrical contact failures through systematic root-cause analysis across a two-week test cycle.

Hydrogen Fuel Cell RC Vehicle

TKS Innovation Fellow · 2023-24

Designed a hydrogen fuel cell power system, managing power distribution and integration. Won a Google-sponsored pitch competition for the technical proposal.

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